Zain Dana Harper

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Graphics and Rendering | Developer Tools | Languages

Summary

Self-directed graphics and tools programmer moving from eleven years of arboriculture; the career change began June 2026, with no professional programming role yet. Builds renderers, shader presets and compilers that check each fast path against a reference.

Technical Skills

C++23, Rust, Python, HLSL, WGSL, TypeScript; D3D11, D3D12, Vulkan, WebGPU; CPU reference rendering and GPU parity checks; color science (Oklab, CAM16, ICC profiles); compilers, type inference, language servers; CMake, GitHub Actions, pytest.

Self-Directed Projects

About 13 years self-directed | Career transition began June 2026

- Built raw-native, a C++23 renderer with a CPU reference beside D3D12 and WebGPU backends; release 0.8.0 passed 60 of 60 tests with MSVC and GCC, and its default renders were byte-identical across Windows, D3D12, Linux and WebAssembly builds.
- Added a Vulkan 1.3 backend to raw-native, merged and not yet released, whose 600 presented frames match the offscreen and D3D12 WARP renders hash for hash.
- Released Elder ENB, a lighting and post-processing preset for Skyrim Special Edition with 38 atmosphere presets: 996,649 downloads and 2,413 endorsements on Nexus Mods, read October 23, 2026.
- Developed BuildLang, a Rust compiler with typed capability effects, type inference, C and shader output and a language server; version 1.10.0 is on crates.io.
- Brought Argonaut's 1990s BRender engine to 64-bit Windows and Linux in Brender Archival, and built Calibrate Pro (ICC profiles, 3D LUTs, DDC/CI) and Build Color (Oklab, CAM16, Delta E), both on PyPI.

Freelance Technical Writer | Independent projects | Began 2017

- Wrote implementation guides, API documentation, and release notes for developers and customers.

Employment

Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Turned site assessments into estimates, proposals, and workable job scopes; coordinated clients, crews, and vendors so teams understood access constraints, timing, and hazards.

Technical Networking Support | Stream/Convergys | 2014 to 2015

Xbox/Microsoft product contract | Wilsonville, Oregon

- Resolved Xbox connectivity problems across TCP/IP, DNS, NAT, firewalls, and routers; documented troubleshooting steps for consistent support handoffs.

Open-Source Contributions: 37 Merged in 30 Repositories

- TOMLKit #549: preserved sibling keys in dotted-key replacement, preventing silent configuration loss. Merged.
- Datasette #2815: corrected malformed composite-key requests from HTTP 500 to HTTP 400, with tests. Merged.

Credentials, Training, and Education

Security, AI and cloud credentials from Qualys, Microsoft, Cisco and Google are listed in the full record: harperz9.github.io/cv.html.

High School Diploma | Wilsonville High School, Oregon | 2013